Course Syllabus

Edit

Instructor:

- Prof. Reza Ebadi (Send emails directly. Please don't send a message via Canvas)
- **rebadi@wpi.edu (mailto:aebadi@wpi.edu)**

Grader:

- Alex Stepanov
- **astepanov@wpi.edu (mailto:aebadi@wpi.edu)**

Location and Time:

- Online
- Lectures will be recorded and posted via Echo 360.

Office Hours:

- By appointment via Zoom

Required Text:

- **Textbook**
  - You **must** purchase the required textbook: Engineering Design with SolidWorks **2025**, David C. Planchard, SDC Publications. (ISBN-13: 978-1-63057-700-2, Find and purchase it from **here (https://wpi.bncollege.com/course-material/course-finder)**).
  - If you purchase the ebook, you can read it through **yuzu.com (https://www.yuzu.com/)** . Search your inbox for instructions and activation code.

Course Objectives:

1. Inspire curiosity about product development

2. Develop three-dimensional visualization skills (2D <=> 3D)
3. Learn to see products from the multiple perspectives of user, designer, manufacturer
4. Develop proficiency with SOLIDWORKS as a powerful Computer Aided Design (CAD) software suite.

AI Use Policy:

- **The features are simple. The mastery is not.** Extrude, cut, revolve, etc. are easy to learn. But the proficiency to envision how simple features combine to create complex objects requires practice. There is no shortcut.
- **This course builds foundation for what comes next.** CAD skills appear in every design course, MQP, and your career. If you don't develop fluency now, you'll struggle to communicate your ideas later.
- **Your future employers expect you to verify AI,** not trust it blindly. Engineers are responsible for their work. "The AI created this part for me" is not a professional defense.
- **How to build proficiency:**
    - Spend the time. 3D visualization develops through repetition, not shortcuts.
    - If stuck, ask AI to explain a feature or workflow; not to think for you.
    - Use AI to learn features we don't cover, but do the modeling yourself.
    - If you can't rebuild it from scratch, you didn't learn it.

You can outsource your thinking to AI, you cannot outsource your understanding.

Grading:

- Labs and Assignments (40%)
- Quizzes and Exams (20%)
- Projects (40%)
- Final Grade Letter:
    - A: ≥ 90
    - B: ≥ 80
    - C: ≥ 70
    - NR: <70

Course Policies and Notes:

- Lab and Homework assignments must be submitted through Canvas.

- Late homework and labs will not be accepted for any reason unless you have made prior arrangements with the instructor.
- Late projects will be accepted with a penalty of 5 points (out of 100) per day.

Academic Honesty:

- Ethical behavior is essential for the engineering professionals due to the nature of the work and is expected during your academic training as well. You are required to comply with all **University policies (https://www.wpi.edu/sites/default/files/docs/Offices/Marketing-Communications/AIStudentguide2019-2020-pages.pdf)** regarding Academic Honesty and to familiarize yourself with those policies.
- Collaborating on assignments is acceptable in this course to support peer-to-peer learning. HOWEVER, you should only discuss the problems conceptually and refrain from exchanging digital copies of the assignments. Let me know if you have any questions about what constitutes academic dishonesty in this course.

Accessibility Services for Students:

- WPI is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must contact the Office of **Accessibility Services (https://www.wpi.edu/offices/office-accessibility-service)** and get them send me proper accommodation letters.

Emotional or Mental Health Distress:

Your academic success in this course is very important to me. If, during the term, you find emotional or mental health issues are affecting that success, please contact WPI's S**tudent Development & Counseling Center (https://www.wpi.edu/offices/student-development-counseling-center)** , which provides counseling appointments and other mental health services.

Netiquette Guidelines:

- Netiquette is the socially and professionally acceptable way to communicate on the Internet. Please abide by these guidelines of "netiquette" when using online communication tools with your classmates and instructor.

- In discussion boards and email messages:
  - Identify yourself. Begin messages with a greeting and close with your name.
  - Avoid sarcasm. It can be misinterpreted and cause hurt feelings.
  - Keep the dialogue collegial and professional. Some discussion topics may be controversial.
  - Do not flame -These are outbursts of extreme emotion or opinion. Think twice before you submit a response. You cannot edit or delete your posts once they have been submitted.
  - Do not use offensive language or profanity.
  - Use clear subject lines for your posts.
  - Do not use all caps. It is the online equivalent of YELLING!
  - Be forgiving. Anyone can make a mistake.
- In Zoom or video conferences:
  - Turn on your camera when your network bandwidth and learning space allows. Facial expressions and body language are an important part of communicating.
  - Connect a few minutes early.
  - Remove clutter or personal items around you.
  - Avoid background noise.
  - Consider using a headset to reduce distractions.
  - Keep your device (phone, computer, etc.) on mute unless you are speaking.
  - Speak clearly, but not too loudly.
  - Don't abuse the chat box, keep the conversation respectful and on topic.
  - Remember that a video conference has the same degree of respect as a live class, consider your appearance behavior on camera the same you would in the classroom.

Tentative Course Outline:

- Course introduction, intro to SOLIDWORKS
- Basics of Computer Aided Design
- Basic SOLIDWORKS part modeling features
- Fundamentals of assemblies
- Fundamentals of drawings, dimensioning and tolerance
- Design intent, and intermediate SOLIDWORKS features
- Intermediate SOLIDWORKS assemblies
- Fundamentals of simulation and analysis

Here is a tentative class schedule. **Please note the dates are subject to change.**